BCA309T: Cryptography & Network Security

ACADEMIC SOLUTIONS MANUAL

Assignment 2: Applied Concepts and Objectives of Cryptography

Q1 (Short) Four Fundamental Cryptographic Goals

The four fundamental cryptographic objectives are: **1. Confidentiality** (preventing unauthorized disclosure), **2. Data Integrity** (detecting unauthorized alteration), **3. Authentication** (verifying entity or message origin identity), and **4. Non-Repudiation** (preventing denial of a previously committed action).

Q2 (Short) Confidentiality Defined with Example

Confidentiality is the security objective ensuring that data contents remain completely inaccessible and concealed from unauthorized individuals, entities, or automated processes. *Example:* An enterprise backing up customer databases to the cloud encrypts files with AES-256; even if the storage bucket is exposed publicly, attackers cannot read the raw database entries.

Q3 (Long) Cryptographic Objectives in Detail & Why All Four Are Jointly Essential

- **Confidentiality:** Conceals information. *Scenario:* Electronic health records transmitted between hospital branches are encrypted to preserve patient privacy against ISP eavesdropping.
- **Integrity:** Detects data tampering. *Scenario:* A bank routing transfer must prove that `"$1,000"` was not altered to `"$1,000,000"` during transmission via cryptographic hashes/MACs.
- **Authentication:** Validates entity identities. *Scenario:* A corporate VPN server validates client digital certificates before granting internal subnet access.
- **Non-Repudiation:** Prevents repudiation of origin. *Scenario:* A stock trader orders 5,000 shares using an asymmetric digital signature; they cannot later claim someone else placed the trade.

Need for Joint Implementation: Achieving these in isolation leaves critical security flaws. Confidentiality without integrity allows active attackers to flip bits in ciphertext (malleability attacks) to subvert logic. Integrity without authentication allows an attacker to send forged data with a valid new checksum. A robust cryptosystem requires all four primitives simultaneously.

Q4 (Short) Substitution vs. Transposition Ciphers

In a **Substitution Cipher**, characters or bits are replaced by other symbols while preserving their positional order (e.g., *Caesar Cipher* shifting letters by 3 positions). In a **Transposition Cipher**, character identities remain unchanged, but their spatial positions are permuted (e.g., *Rail Fence Cipher* writing characters diagonally and reading across rows).

Q5 (Long) Evolution: Classical Ciphers to Modern Standards (DES → AES, RSA, SHA)

Classical ciphers operated on alphabets with pen-and-paper operations; they offered zero security once computing power emerged. The milestones shaping modern cryptography include:

- **DES Standardization (1977):** First standard symmetric cipher. With 56-bit keys and 64-bit blocks, it succumbed to Moore's law and distributed brute-force exhaustion by the late 1990s.
- **AES Transition (2001):** NIST established the Rijndael cipher as AES. Utilizing 128-bit blocks, Substitution-Permutation Networks (SPN), and 128/192/256-bit keys, it resists all known analytical and brute-force attacks today.
- **RSA & Public Key Paradigm (1977):** Enabled widespread open communications without requiring a shared secret key prior to contact, resolving key management scaling.
- **Secure Hash Standards (MD5 → SHA-1 → SHA-2/3):** Collision attacks destroyed MD5 and SHA-1, necessitating SHA-256 and SHA-3 (Keccak) to protect data integrity against preimage and birthday attacks.

Q6 (Short) Cryptographic Key & Importance of Key Management

A **cryptographic key** is a parameterized piece of information (bit string) that controls the mathematical output of a cryptographic algorithm. As stated by Kerckhoffs's Principle, algorithm design should be public; all security rests entirely on key secrecy. **Key management** (generation, distribution, storage, rotation, and revocation) is vital because compromised keys expose all protected traffic regardless of algorithm strength.

Q7 (Long) Role of Cryptography in Network Protection

Network infrastructures expose data packets to intermediate routers, switches, and untrusted airwaves. Cryptography mitigates these risks across three domains:

- **In-Transit Data Protection:** TLS 1.3 encrypts application-layer sessions (HTTPS) preventing packet sniffers (e.g., Wireshark) from capturing cleartext passwords or personal sessions.
- **Access Prevention:** Cryptographic challenge-response protocols (Kerberos, WPA3 SAE) prevent unauthorized nodes from joining internal subnets or accessing Wi-Fi frames.
- **Tamper Detection:** Message Authentication Codes (HMAC) and authenticated encryption modes (AES-GCM) attach cryptographic tags to packets, instantly aborting connections if an adversary flips a bit in transit.

Q8 (Short) Passive vs. Active Attacks

Passive Attacks monitor or intercept data without altering system resources or content; the sender and receiver are unaware. *Example:* Packet sniffing or traffic analysis on an open Wi-Fi network. **Active Attacks** involve altering system data, injecting fraudulent packets, or disrupting services. *Example:* Man-in-the-Middle (MitM) session hijacking or replay attacks.

Q9 (Long) Historical Evolution Against Emerging Threats & The Key-Distribution Crisis

Symmetric-only systems suffered from a fatal scaling bottleneck: for n communicating entities, exactly $n(n-1)/2$ unique keys must be established. In a network of 10,000 users, that requires nearly 50 million keys. Furthermore, securely exchanging a symmetric key across an insecure channel without an existing secure channel represented a chicken-and-egg dilemma. In 1976, Diffie and Hellman solved this key-distribution crisis by introducing asymmetric cryptography: each entity possesses a public key and private key. Users freely publish their public key, reducing key management to $2n$ keys. This paradigm shift enabled the global internet economy.

Q10 (Short) Examples of Modern Cryptographic Algorithms

Modern Symmetric Algorithm: *AES (Advanced Encryption Standard)*, operating on 128-bit blocks with 256-bit keys.
Modern Asymmetric Algorithm: *RSA (Rivest-Shamir-Adleman)* with 2048/4096-bit keys, or *ECDSA* (Elliptic Curve Digital Signature Algorithm).

Q11 (Long) Classification: Classical vs. Modern Cryptography

Characteristic	Classical Cryptography	Modern Cryptography
Basic Unit of Operation	Human alphabetic characters (A–Z).	Binary bit strings (0s and 1s).
Core Mechanisms	Manual substitution and transposition tables.	Complex bit permutations, S-boxes, finite field arithmetic.
Reliance on Secrecy	Security through obscurity (algorithm kept secret).	Kerckhoffs's Principle (algorithm public; only key secret).
Computational Basis	Linguistic patterns and manual paper/rotor devices.	Intractable mathematical problems (factoring, discrete log).